

The background is a light gray gradient. It features several realistic water droplets of various sizes, some with highlights and shadows, scattered across the top and bottom edges. Faint, thin white circles are also visible in the background, centered around the text area.

Unit-4

Fuzzy Logic

the basic terminologies and notations

Definition 1: Membership function (and Fuzzy set)

If X is a universe of discourse and $x \in X$, then a fuzzy set A in X is defined as a set of ordered pairs, that is $A = \{(x, \mu_A(x)) | x \in X\}$ where $\mu_A(x)$ is called the **membership function** for the fuzzy set A .

$\mu_A(x)$ map each element of X onto a membership grade (or membership value) between 0 and 1 (both inclusive).

Question: How (and who) decides $\mu_A(x)$ for a fuzzy set A in X ?

the basic terminologies and notations

Example:

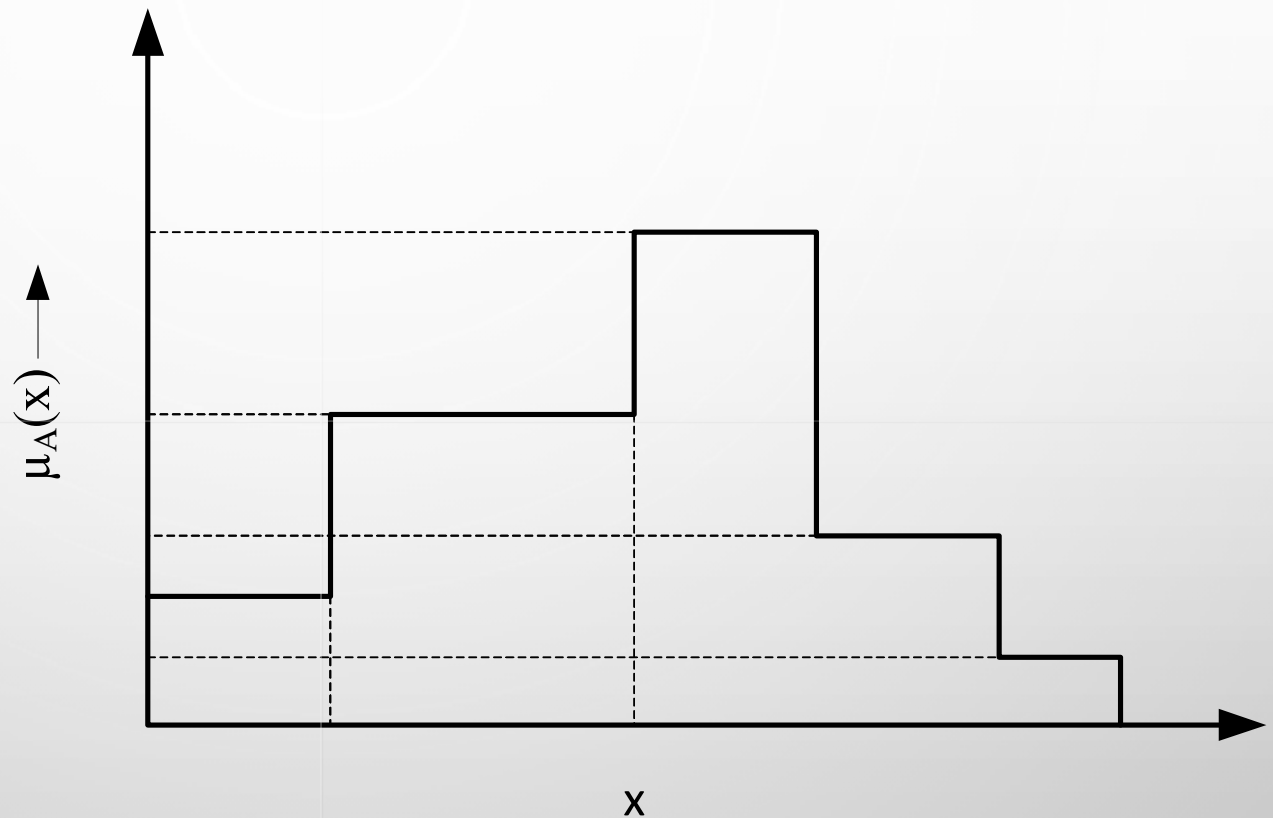
X = All cities in India

A = City of comfort

$A = \{(\text{New Delhi}, 0.7), (\text{Bangalore}, 0.9), (\text{Chennai}, 0.8), (\text{Hyderabad}, 0.6),$
 $\text{Kolkata}, 0.3), (\text{Kharagpur}, 0)\}$

Membership function with discrete membership values

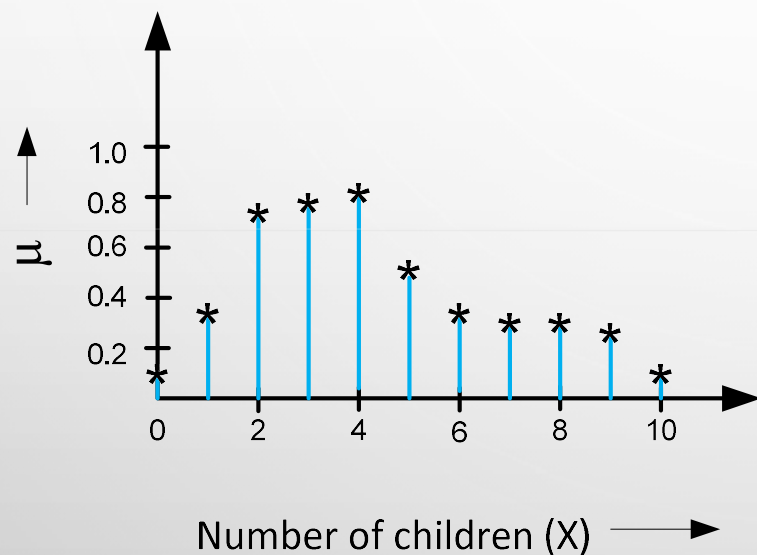
The membership values may be of discrete values.



A fuzzy set with discrete values of μ

Membership function with discrete membership values

Elements or their membership values (or both) also may be of discrete values.



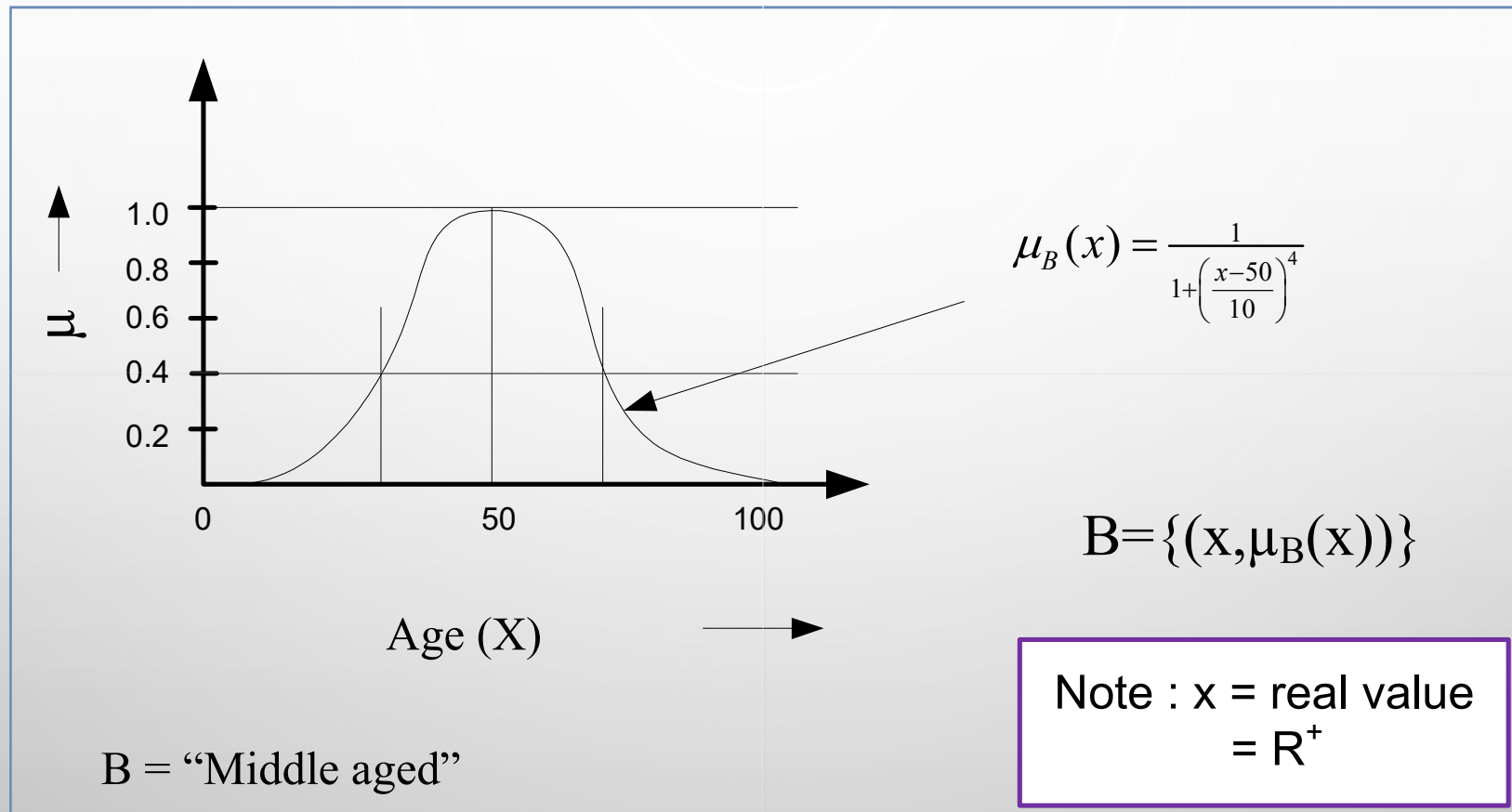
A = "Happy family"

$$A = \{(0, 0.1), (1, 0.30), (2, 0.78), \dots, (10, 0.1)\}$$

Note : X = discrete value

How you measure happiness ??

Membership function with continuous membership values



The background of the slide features a light gray grid pattern. In the top-left corner, there are several realistic water droplets of varying sizes, some with highlights and shadows. In the bottom-right corner, there are more water droplets, including a large, prominent one with a clear highlight. The overall aesthetic is clean and modern.

Fuzzy Membership Functions

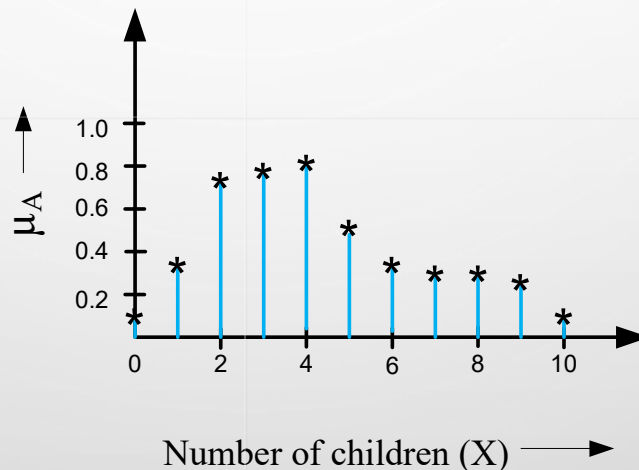
Fuzzy membership functions

A fuzzy set is completely characterized by its membership function (sometimes abbreviated as MF and denoted as μ). So, it would be important to learn how a membership function can be expressed (mathematically or otherwise).

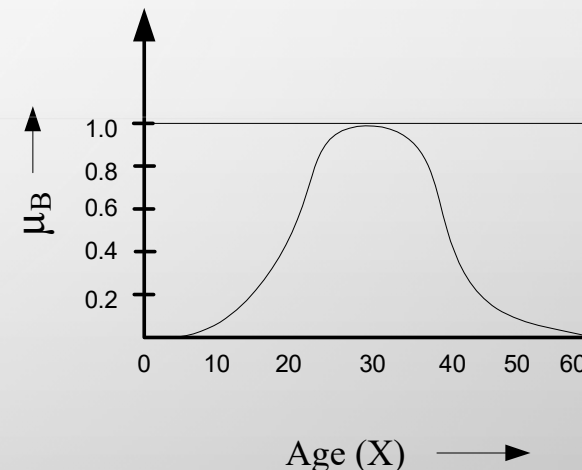
A membership function can be on

- a) a discrete universe of discourse and
- b) a continuous universe of discourse.

Example:



A = Fuzzy set of "Happy family"

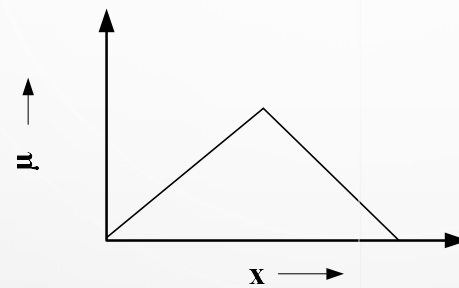


B = "Young age"

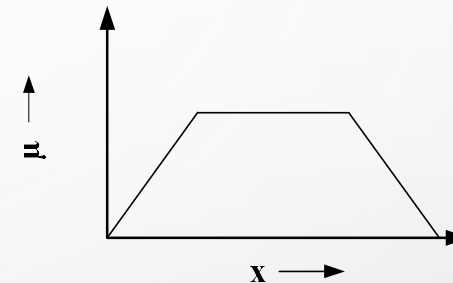
uzzy membership functions

A membership function on a discrete universe of course is trivial. However, a membership function on a continuous universe of discourse needs a special attention.

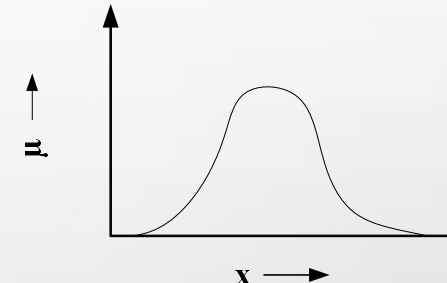
Following figures shows typical examples of membership functions.



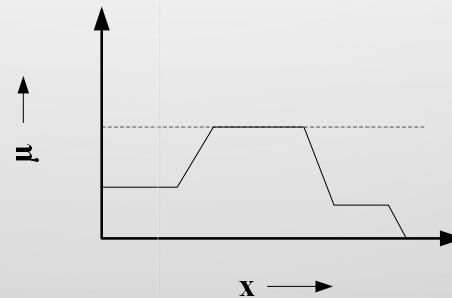
< triangular >



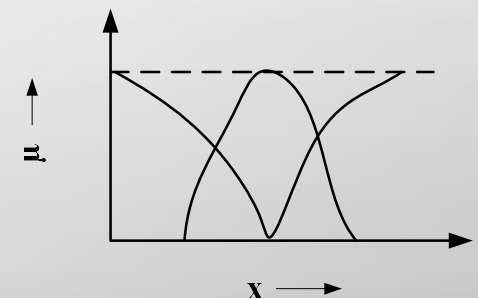
< trapezoidal >



< curve >



< non-uniform >

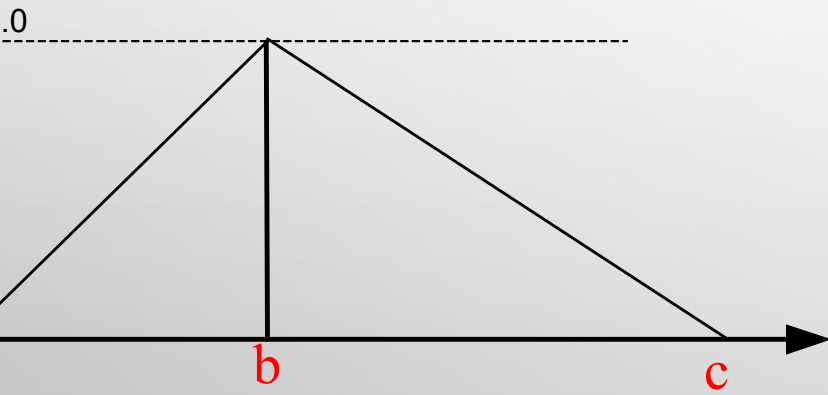


< non-uniform >

Fuzzy MFs : Formulation and parameterization

In the following, we try to parameterize the different MFs on a continuous universe of discourse.

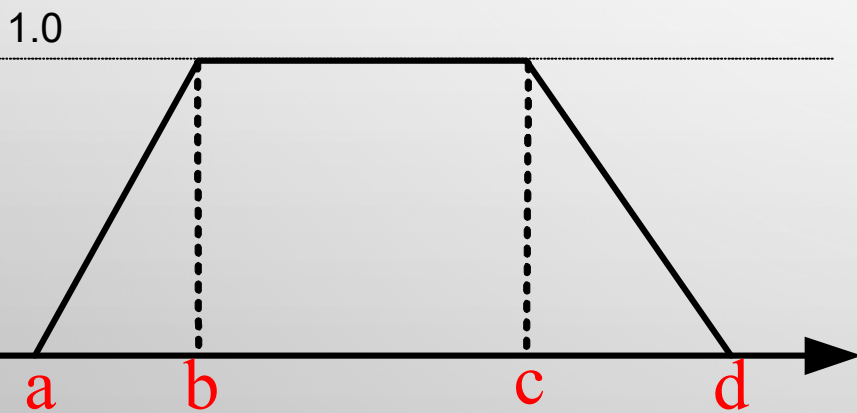
Triangular MFs : A triangular MF is specified by three parameters $\{a, b, c\}$ and can be formulated as follows.



$$\text{triangle}(x; a, b, c) = \begin{cases} 0 & \text{if } x \leq a \\ \frac{x - a}{a - b} & \text{if } a \leq x \leq b \\ \frac{c - x}{c - b} & \text{if } b \leq x \leq c \\ 0 & \text{if } c \leq x \end{cases}$$

Fuzzy MFs: Trapezoidal

Trapezoidal MF is specified by four parameters $\{a, b, c, d\}$ and can be defined as follows:

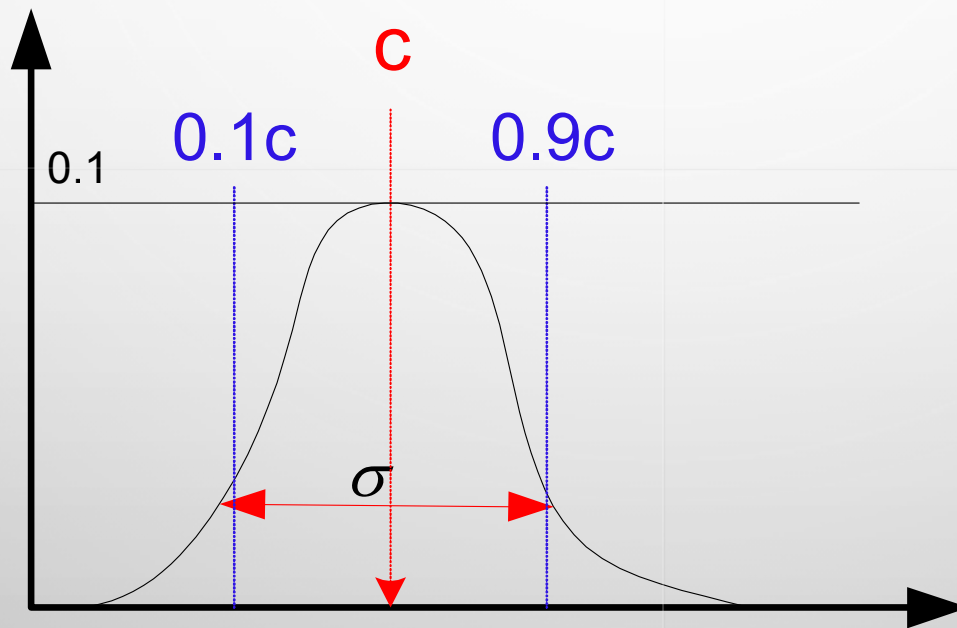


$$\text{trapezoid}(x; a, b, c, d) = \begin{cases} 0 & \text{if } x \leq a \\ \frac{x - a}{b - a} & \text{if } a \leq x \leq b \\ 1 & \text{if } b \leq x \leq c \\ \frac{d - x}{d - c} & \text{if } c \leq x \leq d \\ 0 & \text{if } d \leq x \end{cases}$$

by MFs: Gaussian

Gaussian MF is specified by two parameters $\{c, \sigma\}$ and can be defined as below:

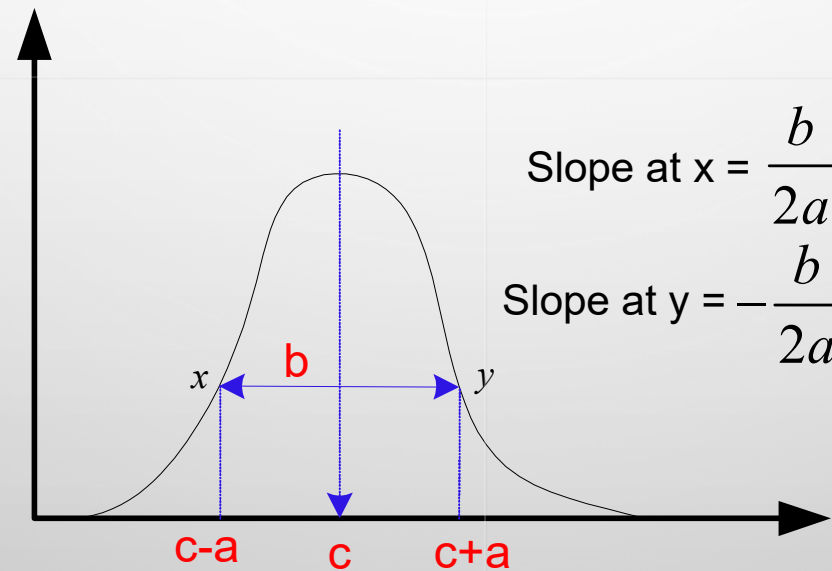
$$\text{gaussian}(x; c, \sigma) = e^{-\frac{1}{2}\left(\frac{x-c}{\sigma}\right)^2}$$



Fuzzy MFs: Generalized bell

also called **Cauchy MF**. A generalized bell MF is specified by three parameters a, b, c and is defined as:

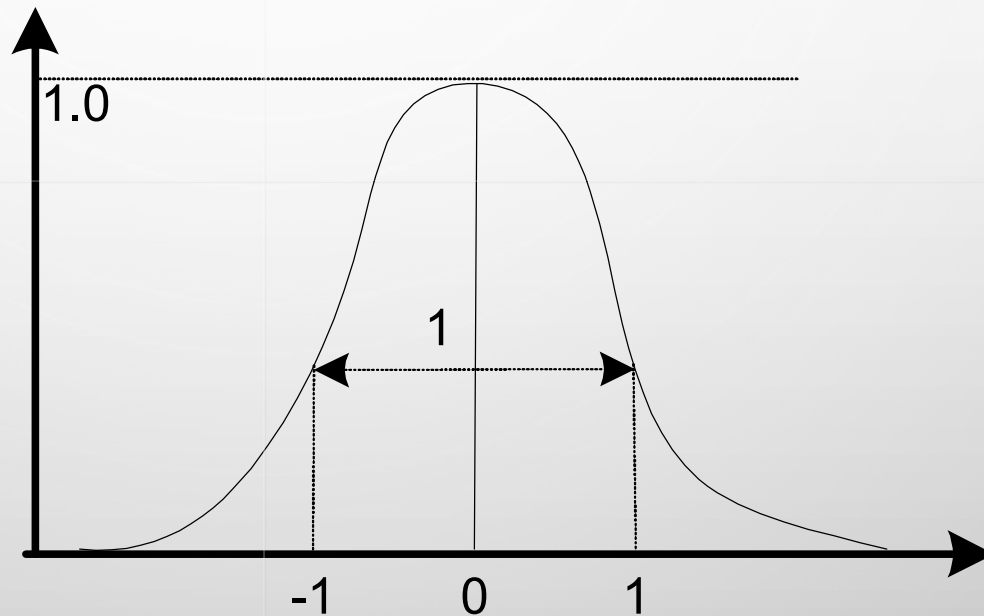
$$bell(x; a, b, c) = \frac{1}{1 + \left| \frac{x - c}{a} \right|^{2b}}$$



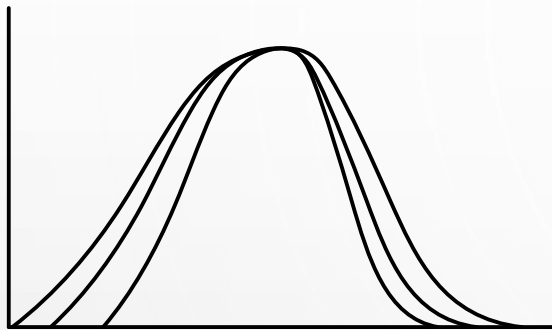
Example: Generalized bell MFs

Example: $\mu(x) = \frac{1}{1+|x|^2}$;

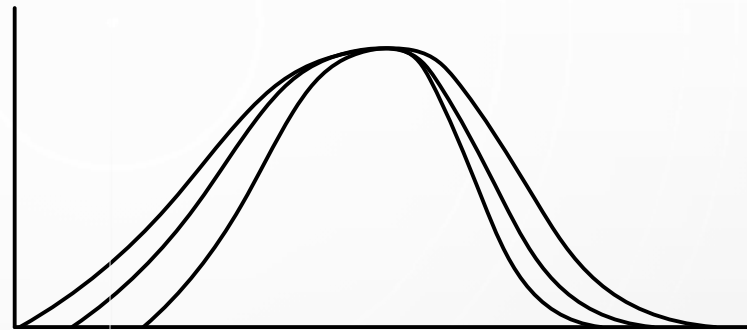
$a = b = 1$ and $c = 0$;



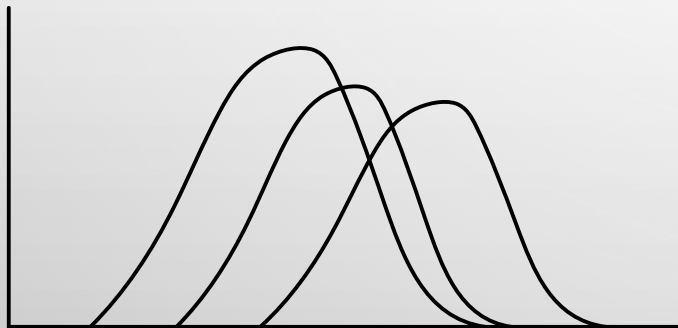
Generalized bell MFs: Different shapes



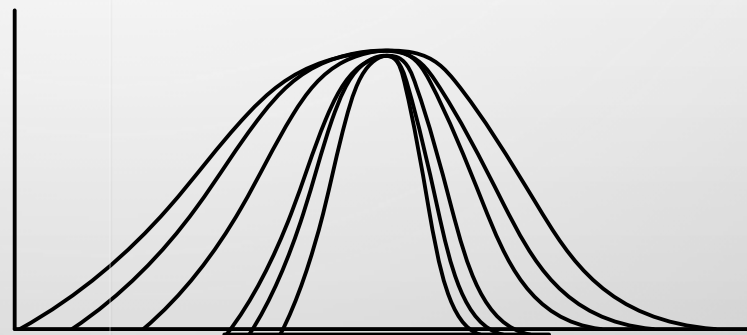
Changing a



Changing b



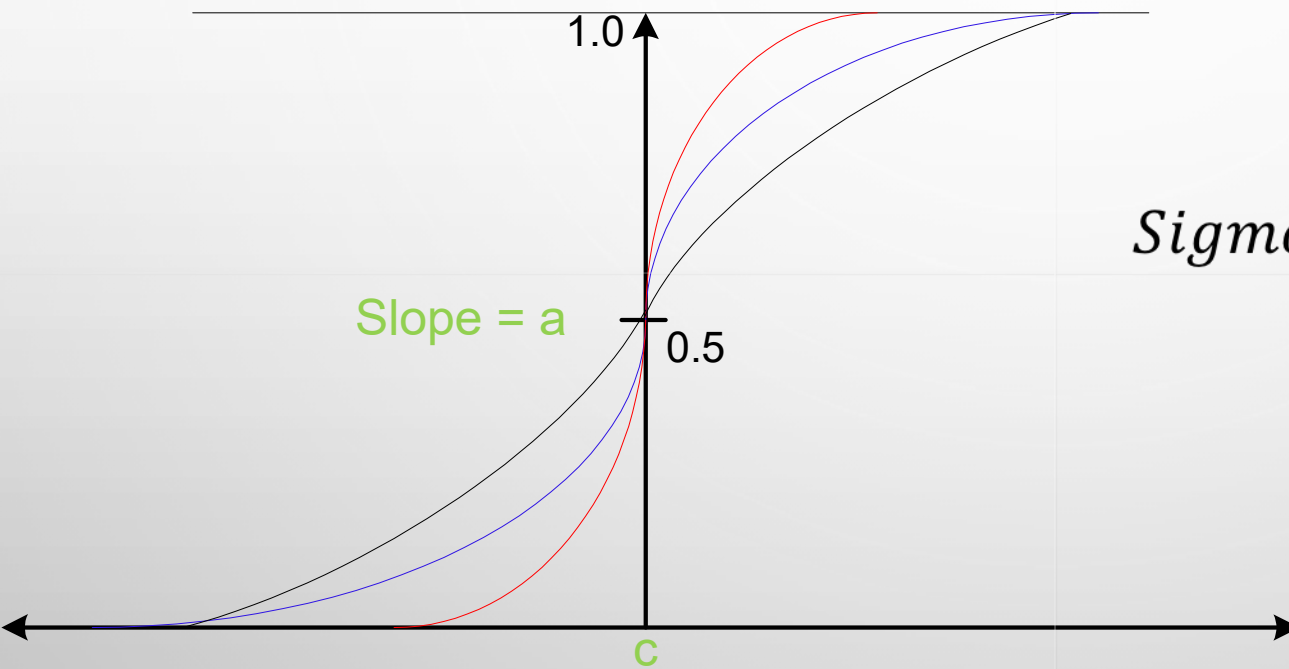
Changing a



Changing a and b

zzy MFs: Sigmoidal MFs

Parameters: $\{a, c\}$; where c = crossover point and a = slope at c ;



$$\text{Sigmoid}(x; a, c) = \frac{1}{1 + e^{-\left[\frac{x-c}{a}\right]}}$$

My MFs : Example

Example : Consider the following grading system for a course.

Excellent = Marks ≤ 90

Very good = $75 \leq \text{Marks} \leq 90$

Good = $60 \leq \text{Marks} \leq 75$

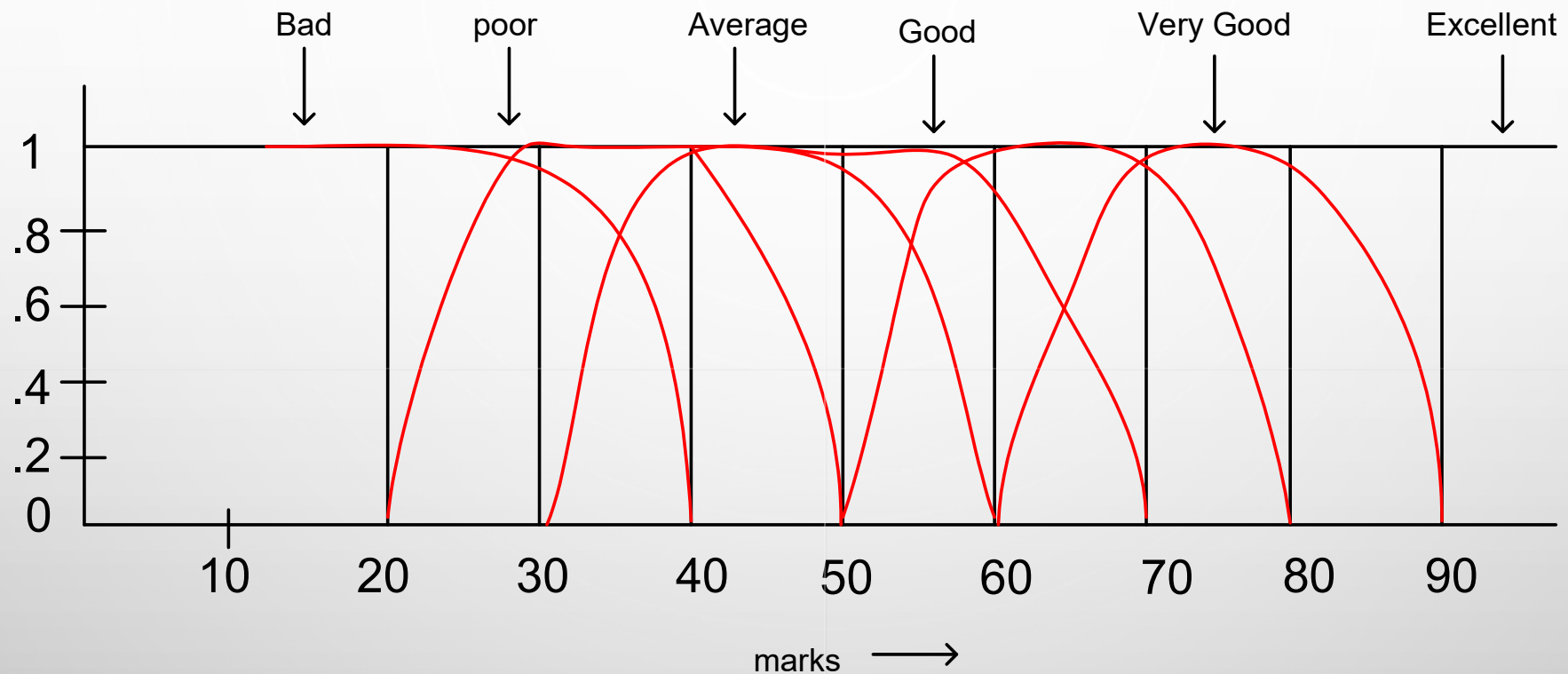
Average = $50 \leq \text{Marks} \leq 60$

Poor = $35 \leq \text{Marks} \leq 50$

Bad = Marks ≤ 35

ding System

A fuzzy implementation will look like the following.



You can decide a standard fuzzy MF for each of the **fuzzy grade**.

Few More on Membership Functions

Generation of MFs

From a membership function of a fuzzy set representing a **linguistic hedge**, we can generate more MFs representing several other linguistic hedges using the concepts of **Concentration** and **Dilation**.

Concentration: $A^k = [\mu_A(x)]^k; k > 1$

Dilation: $A^k = [\mu_A(x)]^k; k < 1$

Example : Age = { Young, Middle-aged, Old }

For example, corresponding to Young, we have : **Not young, Very young, Not very young** and so on. Similarly, with Old we can have : **Not old, Very old, Very very old, Extremely old**, etc.

$\mu_{Extremely\ old}(x) = ((\mu_{Old}(x))^2)^2$ and so on

$\mu_{More\ or\ less\ old}(x) = A^{0.5} = (\mu_{Old}(x))^{0.5}$

Fuzzy variables and values

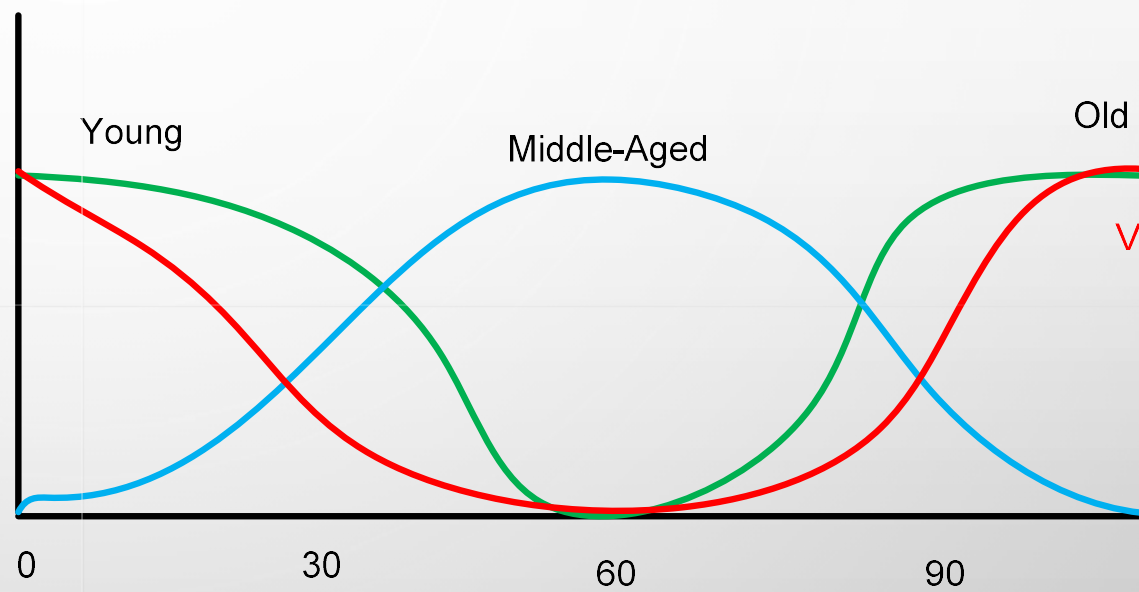
$$\mu_{young}(x) = \text{bell}(x, 20, 2, 0) = \frac{1}{1 + (\frac{x}{20})^4}$$

$$\mu_{middle-aged}(x) = \text{bell}(x, 30, 3, 100) = \frac{1}{1 + (\frac{x-100}{30})^6}$$

$$\mu_{middle-aged}(x) = \text{bell}(x, 30, 60, 50)$$

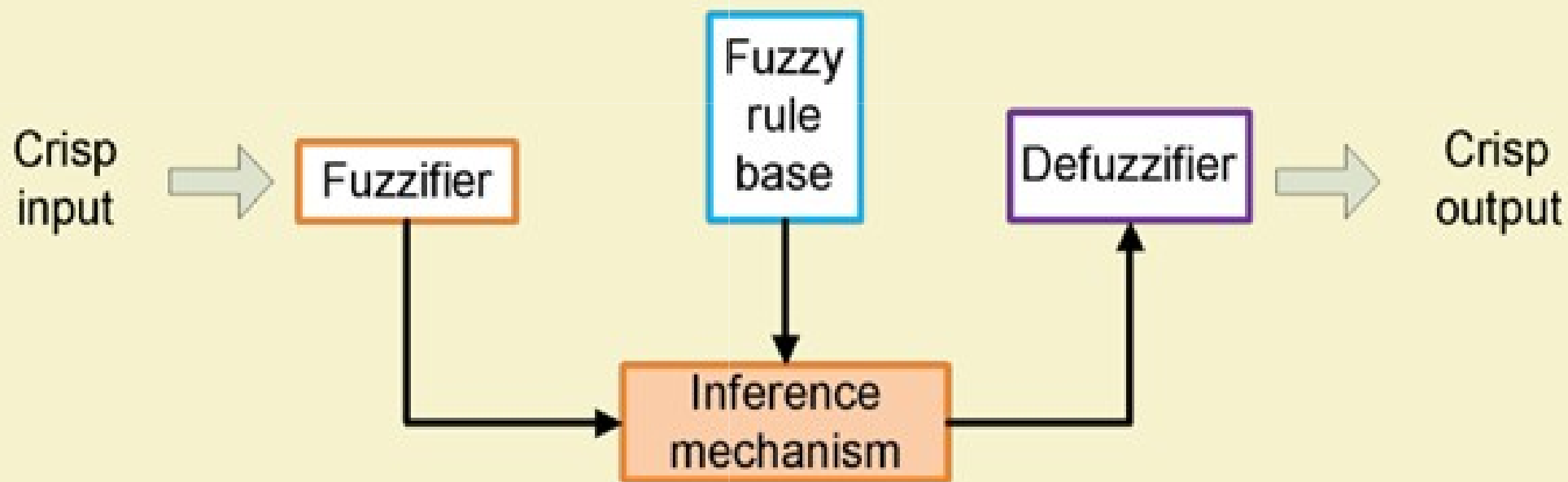
$$\mu_{not\ young} = \overline{\mu_{young}(x)} = 1 - \mu_{young}(x)$$

$$\mu_{young\ but\ not\ too\ young} = \mu_{young}(x) \cap \overline{\mu_{young}(x)}$$



Generic structure of a Fuzzy system

Following figure shows a general framework of a fuzzy system.



What is defuzzification?

- Defuzzification means the fuzzy to crisp conversion.

Example 1.

Suppose, T_{HIGH} denotes a fuzzy set representing **temperature is High**.

T_{HIGH} is given as follows.

$$T_{HIGH} = \{(15, 0.1), (20, 0.4), (25, 0.45), (30, 0.55), (35, 0.65), (40, 0.7), (45, 0.85), (50, 0.9)\}$$

- What is the crisp value that implies the high temperature?

Why defuzzification?

The fuzzy results generated can not be used in an application, where decision has to be taken only on crisp values.

Example.

If temperature is T_{HIGH} Then rotation is R_{FAST} .

Here, may be input T_{HIGH} is fuzzy, but action rotation should be based on the crisp value of R_{FAST} .

Defuzzification methods

A number of defuzzification methods are known. Such as

- 1) **Lambda-cut method**
- 2) **Weighted average method**
- 3) **Maxima methods**
- 4) **Centroid methods**

Lambda-cut for a fuzzy set : Example

$$A_1 = \{(x_1, 0.9), (x_2, 0.5), (x_3, 0.2), (x_4, 0.3)\}$$

$$\lambda = 0.6$$

$$A_{0.6} = \{(x_1, 1), (x_2, 0), (x_3, 0), (x_4, 0)\} = \{x_1\}$$

$$A_2 = \{(x_1, 0.1), (x_2, 0.5), (x_3, 0.8), (x_4, 0.7)\}$$

$$\lambda = 0.2$$

$$A_{0.2} = \{(x_1, 0), (x_2, 1), (x_3, 1), (x_4, 1)\} = \{x_2, x_3, x_4\}$$



Fuzzy Systems : Fuzzy Logic Controller

Concept of fuzzy theory can be applied in many applications, such as fuzzy reasoning, fuzzy clustering, fuzzy programming, etc.

Out of all these applications, **fuzzy reasoning**, also called "**fuzzy logic controller (FLC)**" is an important application.

Fuzzy logic controllers are special expert systems. In general, a FLC employs a knowledge base expressed in terms of a **fuzzy inference rules** and a **fuzzy inference engine** to solve a problem.





Fuzzy Systems : Fuzzy Logic Controller

We use FLC where an exact mathematical formulation of the problem is possible or very difficult.

These difficulties are due to non-linearities, time-varying nature of process, large unpredictable environment disturbances, etc.

Fuzzy Systems : Fuzzy Logic Controller

A general fuzzy controller consists of four modules:

- a fuzzy rule base,
- a fuzzy inference engine,
- a fuzzification module, and
- a defuzzification module.

Fuzzy Systems : Fuzzy Logic Controller

Fuzzy Controller

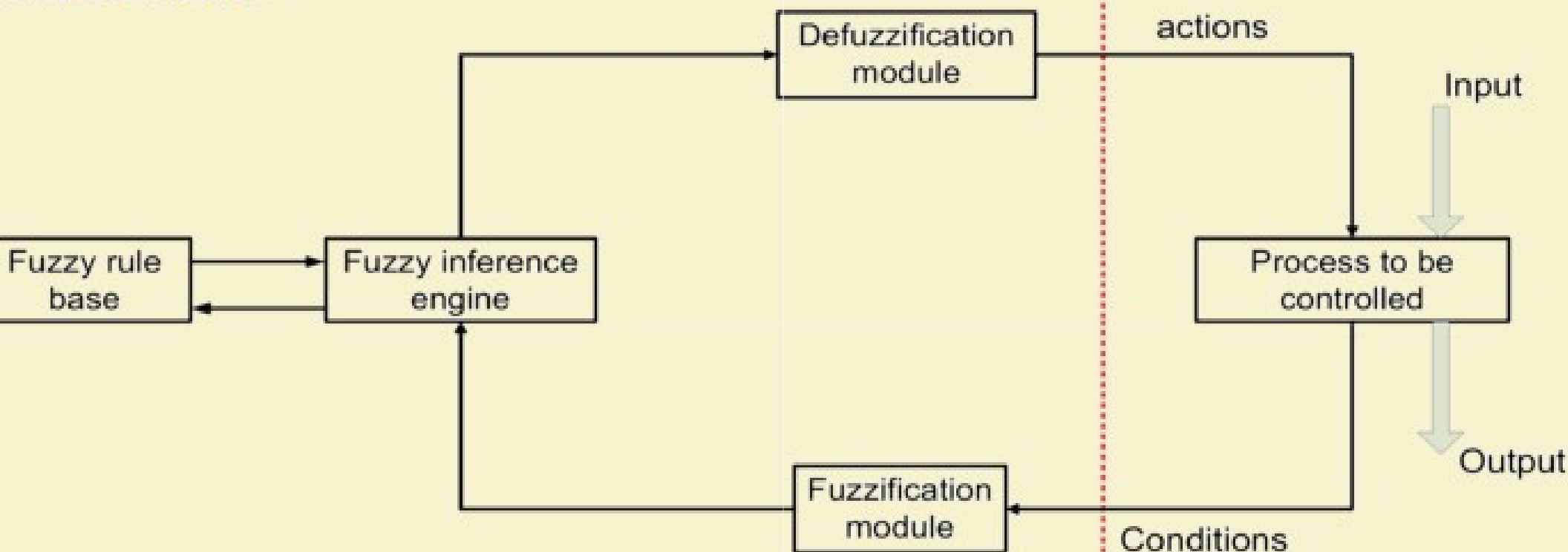


Figure 1: A general scheme of a fuzzy controller

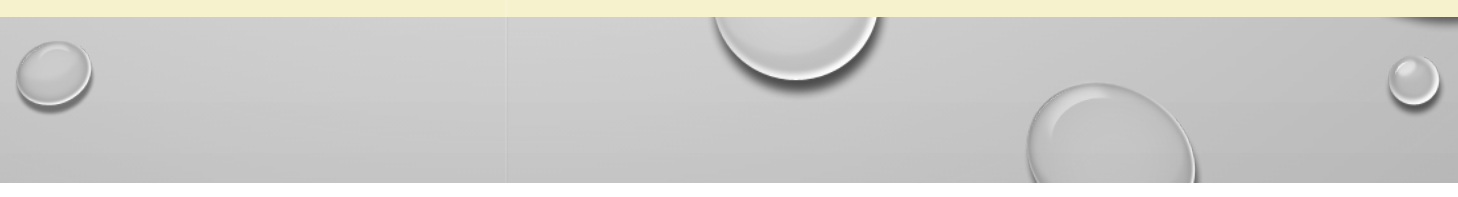


Fuzzy Systems : Fuzzy Logic Controller

As shown in **Figure 1**, a fuzzy controller operates by repeating a cycle of the following **four steps** :

Step 1: **Measurements** (inputs) are taken of all variables that represent the relevant condition of controller process.

Step 2: These measurements are converted into appropriate fuzzy sets to express measurements uncertainties. This step is called **fuzzification**.



Fuzzy Systems : Fuzzy Logic Controller

Step 3: The fuzzified measurements are then used by the inference engine to evaluate the control rules stored in the fuzzy rule base. The result of the evaluation is a **fuzzy set** (or several fuzzy sets) defined on the universe of possible actions.

Step 4: This output fuzzy set is then converted into a single (crisp) value (or a vector of values). This is the final step called **defuzzification**. The defuzzified values represent actions to be taken by the fuzzy controller.